EE/CS 148B - Homework 3

Aiden Di Carlo

June 1, 2026

Vision Transformer

(a) vit_pooling: For tasks requiring spatial reasoning, attention pooling is expected to perform best because it can learn to weight tokens selectively based on task-relevant spatial cues, rather than treating all positions equally (mean pooling) or relying on a single generic summary (CLS). Mean pooling discards per-patch identity by averaging, which collapses distinctions between, say, a red cube on the left versus the right; the CLS token suffers the same problem more acutely, since it is trained to summarize global semantics and does not preserve fine-grained spatial layout at all. When an entire image is condensed into a single CLS vector before being passed to a language model, all explicit positional relationships between objects are lost. The model receives no direct signal about where each object is, only a holistic blend of what is in the image.

(b) vit_patch_size:

(1) Patch counts and attention cost. For a 224×224 image the number of patches is $N = (224/P)^2$:

P	N
8	784
16	196
32	49

Self-attention scales as $O(N^2 d_{\text{model}})$, and $N \propto P^{-2}$. Thus, the overall cost scales as $O(P^{-4} d_{\text{model}})$. Halving the patch size increases attention compute by $2^4 = 16\times$, so shrinking P is extremely expensive.

(2) Wall-clock times (batch of 16 images, $d_{\text{model}} = 384$, 6 heads, 6 blocks; RTX 4070 Laptop GPU; 5 warmup + 20 timed steps):

P	Mean (ms)	Std (ms)
8	84.9	1.8
16	14.8	1.3
32	10.1	1.3

The empirical ratios ($8.4\times$ between $P = 8$ and $P = 16$, and $1.5\times$ between $P = 16$ and $P = 32$) are smaller than the theoretical $16\times$ and $256\times$ because the GPU is compute-bound only for the largest sequence length; at $P = 16$ and $P = 32$ the bottleneck shifts to memory bandwidth and kernel launch overhead, compressing the observed differences. The $P = 8$ overhead, roughly $8\times$ slower than $P = 16$, confirms that the quadratic attention term dominates at high patch counts.

(3) Accept smaller patches when fine-grained spatial fidelity is critical and the compute budget allows it (e.g., OCR, dense prediction, or counting small objects where coarser tokens would merge distinct features).

CLIP-Style Contrastive Pretraining

(a) infonce: The loss is averaged in both directions because each (image, text) pair should serve as a positive for both retrieval directions—finding the right caption given an image, and the right image given a caption. Using only one direction would bias the model toward one retrieval mode and leave half the pairwise supervision signal in each batch unused.

(b) clip_train:

Validation accuracy rises steeply during the first eight epochs (53% → 87%), then continues to improve at a diminishing rate before plateauing near 91% from epoch 17 onward. Training loss, by contrast, decreases monotonically throughout all 20 epochs—including the final epochs where accuracy is essentially flat—indicating the model keeps tightening image–text alignment on training pairs past the point of diminishing returns for zero-shot classification. This is expected under EuroSAT's class-template captions: many examples per batch share the same caption, so the loss can still improve by better separating near-duplicate positives even after the coarser class-level decision boundaries are learned.

(c) clip_zeroshot:

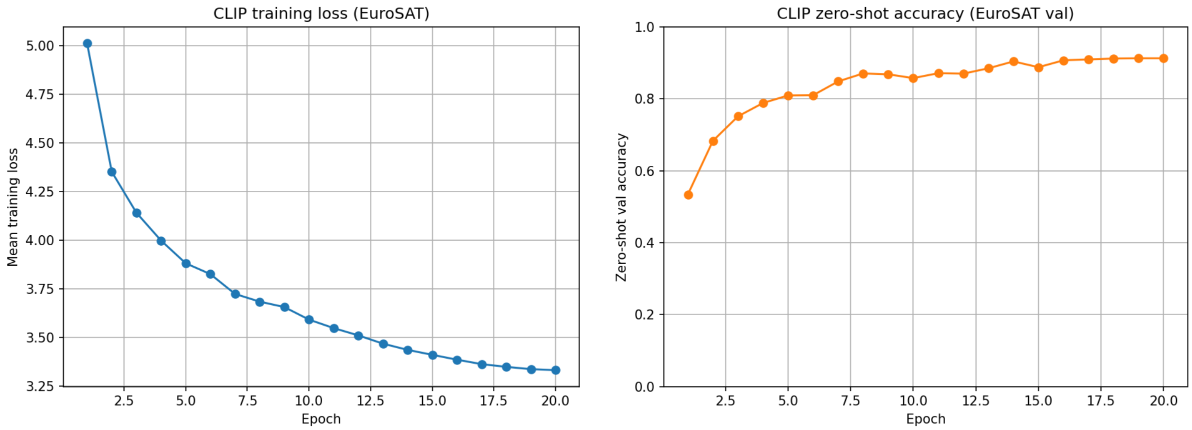

Figure 1: Left: mean training loss per epoch. Right: zero-shot validation accuracy per epoch. EuroSAT, 20 epochs, A100 GPU.

Nearly all errors are reasonable confusions between visually similar classes: the four agricultural/vegetation categories (Pasture, Herbaceous Vegetation, Annual Crop, Permanent Crop) appear as low-texture green patches at 64×64 resolution, making them hard to distinguish, and Industrial Buildings versus Residential Buildings are both regular grids of rooftops separated by roads. The one surprising error is Annual Crop predicted as Highway: at this resolution, the parallel furrows of a ploughed field closely resemble lane markings on a multi-lane road. Because all five wrong predictions still place the correct class in the top-3, the mistakes reflect genuine local ambiguity rather than a failure of the embedding space: the model has learned semantically coherent clusters ("green land," "built environment," "water") but struggles to separate fine-grained classes within each cluster.